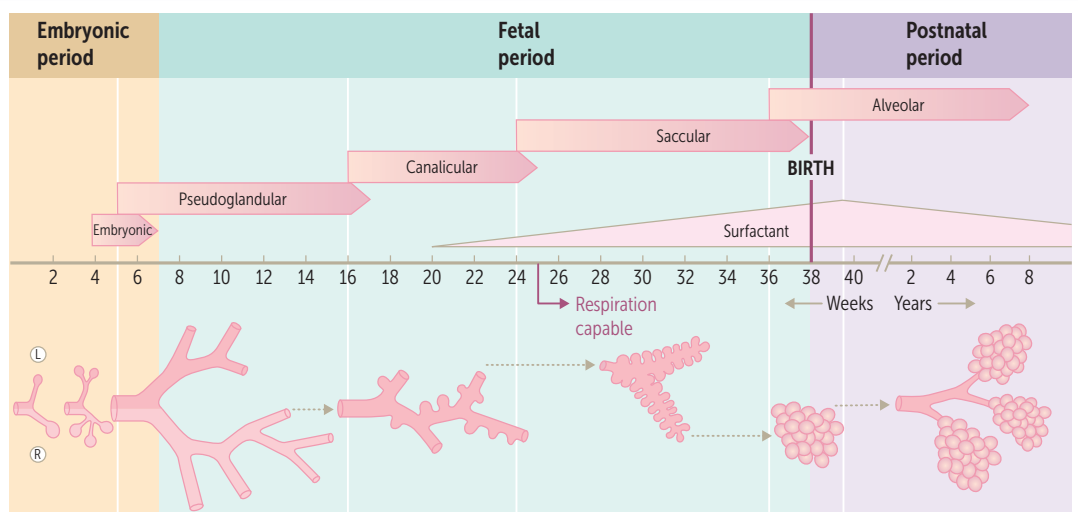


► RESPIRATORY—EMBRYOLOGY

Lung development

Occurs in five stages. Begins with the formation of lung bud from distal end of respiratory diverticulum during week 4 of development. Every pulmonologist can see alveoli.

STAGE	STRUCTURAL DEVELOPMENT	NOTES
Embryonic (weeks 4–7)	Lung bud → trachea → bronchial buds → mainstem bronchi → secondary (lobar) bronchi → tertiary (segmental) bronchi.	Errors at this stage can lead to tracheoesophageal fistula, tracheal atresia/stenosis and pulmonary agenesis.
Pseudoglandular (weeks 5–17)	Endodermal tubules → terminal bronchioles. Surrounded by modest capillary network.	Respiration impossible, incompatible with life.
Canalicular (weeks 16–25)	Terminal bronchioles → respiratory bronchioles → alveolar ducts. Surrounded by prominent capillary network.	Airways increase in diameter. Pneumocytes develop starting at week 20 (surfactant synthesis). Respiration capable at week 25.
Saccular (week 24–birth)	Alveolar ducts → terminal sacs. Terminal sacs separated by 1° septae.	
Alveolar (week 36–8 years)	Terminal sacs → adult alveoli (due to 2° septation).	In utero, “breathing” occurs via aspiration and expulsion of amniotic fluid → ↑ pulmonary vascular resistance through gestation. At birth, air replaces fluid → ↓ pulmonary vascular resistance.

**Choanal atresia**

Blockage of posterior nasal opening. Often associated with bony abnormalities of the midface. Most often unilateral. When bilateral, is an emergency and presents with upper airway obstruction, noisy breathing, and/or cyanosis that worsens during feeding and improves with crying. Diagnosed by failure to pass nasogastric tube (beyond the pharynx) and confirmed with CT scan.

Often part of multiple malformation syndromes, such as **CHARGE** syndrome:

- **C**oloboma of eye
- **H**ear defects
- **A**tresia of choanae
- **R**estricted growth and development
- **G**enitourinary defects
- **E**ar defects

Lung malformations

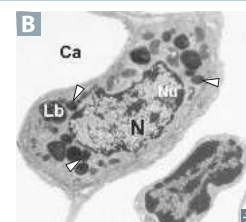
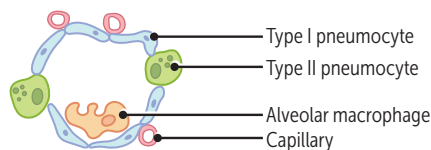
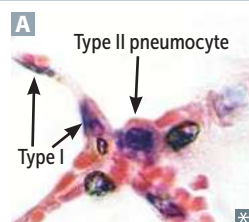
Pulmonary hypoplasia	Poorly developed bronchial tree with abnormal histology. Associated with congenital diaphragmatic hernia (usually left-sided), bilateral renal agenesis (Potter sequence).
Bronchogenic cysts	Caused by abnormal budding of the foregut and dilation of terminal or large bronchi. Discrete, round, sharply defined, fluid-filled densities on CXR (air-filled if infected). Generally asymptomatic but can drain poorly → airway compression, recurrent respiratory infections.

Club cells

Microciliated; low columnar/cuboidal with secretory granules. Located in bronchioles. Degrade toxins (P-450); secrete surfactant-like component; progenitor for club and ciliated cells.

Alveolar cell types

Type I pneumocytes	Squamous. 97% of alveolar surfaces. Thinly line the alveoli A for optimal gas exchange.	Pores of Kohn —anatomical communications between alveoli that allow for passing of air, fluid, phagocytes, and bacteria (in pneumonia).
Type II pneumocytes	Cuboidal and clustered B . 2 functions: 1. Serve as stem cell precursors for 2 cell types (type I and type II pneumocytes); proliferate during lung damage. 2. Secrete surfactant from lamellar bodies (arrowheads in B). Application of Law of Laplace in alveoli—alveoli have ↑ tendency to collapse on expiration as radius ↓.	Surfactant —↓ alveolar surface tension, ↓ alveolar collapse, ↓ lung recoil, and ↑ compliance. Composed of multiple lecithins, mainly dipalmitoylphosphatidylcholine (DPPC). Synthesis begins ~20 weeks of gestation and achieves mature levels ~35 weeks of gestation. Glucocorticoids are important for fetal surfactant synthesis and lung development. Collapsing pressure = $2 \text{ (surface tension)/radius}$
Alveolar macrophages (dust cells)	Phagocytose foreign materials; release cytokines and alveolar proteases.	Hemosiderin-laden macrophages may be found (eg, pulmonary edema, alveolar hemorrhage).

**Neonatal respiratory distress syndrome****A**

Surfactant deficiency → ↑ surface tension → alveolar collapse (“ground-glass” appearance of lung fields) **A**.

Risk factors: prematurity, diabetes during pregnancy (due to ↑ fetal insulin), C-section delivery (↓ release of fetal glucocorticoids; less stressful than vaginal delivery).

Treatment: maternal glucocorticoids before birth; exogenous surfactant for infant.

Therapeutic supplemental O₂ → **R**etinopathy of prematurity, **I**ntraventricular hemorrhage, **B**ronchopulmonary dysplasia (**RIB**).

Persistently low O₂ tension → risk of PDA.

Screening tests for fetal lung maturity: lamellar body count and lecithin-sphingomyelin (L/S) ratio in amniotic fluid (≥ 2 is healthy; < 1.5 predictive of NRDS), foam stability index, surfactant-albumin ratio.

